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Report on Geomagnetic Field Reconstruction using Iterative Regularisation Techniques

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Abstract

The existence of the Southern Atlantic Anomaly (SAA) in the geomagnetic field remains a topic of debate within the literature. We present reconstructions of the radial geomagnetic field at the Core-Mantle Boundary (CMB) which to high certainty support findings of a region of reversed flux. The employed model solves the inverse problem of determining the Gauss coefficients of a spherical harmonic expansion using data from ESA's Swarm mission. Two inversions are carried out, one with unweighted data in the L_2-norm and the other with robustly reweighted data in the L_1-norm, with the latter model determined using the Iteratively Reweighted Least Squares algorithm with Huber weights and the perturbed L_p-norm scheme due to Ekblom. Both of the produced models reveal flux reversal beneath the Southern Atlantic Ocean. Subsequent analysis of the prediction errors and the spatial resolution of the model leads to the conclusion that the null-hypothesis of purely dipolar geomagnetism must be rejected.

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Contents

Abstract 1
Artificial Intelligence Declaration i
1 Introduction 1
2 Data and Experimental Design 1
3 Theory — Physics and Forward Model 1
4 Theory — Mathematical Treatment of the Inverse Problem 3
4.1 Least Squares Solution 3
4.2 Stability Analysis 4
4.3 Regularisation Theory 4
4.4 Robust Iteratively Reweighted Least Squares Algorithm 5
4.5 Covariance and Resolution Matrices 6
5 Results 8
5.1 Unregularised Least Squares Solution 8
5.2 Non-Robust L_2-Regularised Solution 8
5.3 Robust L_1-Regularised Solution 11
5.3.a Covariance and Resolution Characteristics 12
6 Discussion 14
7 Conclusion 14
8 Bibliography 16
A Appendix 17
A.1 Large Plot of Solution to L_2-regularised Problem 17
A.2 Large Plot of Solution to Robust L_1-regularised Problem 17
A.3 Implementation of Robust Iteratively Reweighted Least Squares Algorithm 18

Artificial Intelligence Declaration

The body of work presented below is solely authored by the listed authors of this document, Jeppe Klitgaard and Mingming Li, and none of the presented material has been produced by means of *Generative AI* in the manner requiring citation as described by [1].

Generative AI has, however, been used selectively in the generation of limited parts of the code. GitHub Copilot tab-completion has been used during the development of the attached code and in the generation of an initial version of the following, which is also clearly sign-posted by comments in the attached code:

- Initial version of the plotting code for the residual plot
- Alternative L-curve corner finding algorithm: Normalised Menger Curvature
- Alternative L-curve corner finding algorithm: Chord Distance

The generated code output has in all instances been checked and often revised by the authors and is *not* considered to be a *significant* contribution to total intellectual work undertaken and presented.

1 Introduction

This report seeks to determine the radial magnetic field, B_r, at the Core-Mantle Boundary (CMB) of the Earth using data collected as part of the three-satellite ESA Swarm mission. This may be considered as an *inverse problem* in which the model space is spanned by the Gauss coefficients of the expansion of the projection onto the spherical harmonics.

A derivation of the forward model is presented using the Ampère-Maxwell law and the assumption of interior sources after which the relevant theory from regularisation theory and the study of inverse problems is laid out. The inversion, which produces a model from a set of observations, is proposed to be carried out using either L_2 or L_1 regularisation, the latter of which is solved using the proposed Iteratively Reweighted Least Squares (IRLS) algorithm with robust weights.

Suitable candidates for the regularisation parameters, α^2, are determined by grid searches and the familiar L-curve heuristic after which the resulting models are used to reconstruct predictions of the radial field at the CMB. Estimates of the model uncertainties and their resolution are carried out alongside an estimated prediction confidence found using the plug-in principle and the assumption of univariate, uncorrelated noise.

The results in the form of model projections onto the CMB and associated power spectra are then evaluated against the dipole model of the geomagnetic field, which serves as the null-hypothesis. The outcomes of both the L_2 and robust L_1-regularised models are compared with particular consideration of the bias introduced by the regularisation terms.

Lastly, the findings of the report are summarised and the conclusion is presented.

2 Data and Experimental Design

The dataset used in this report consists of $N_d = 7561$ globally distributed measurements obtained from the three satellites of the Swarm mission. Each observation consists of the measured radial component of the Earth's magnetic field alongside the satellite location at the time of measurement in the form of the spherical coordinates (r, ϑ, λ), where r is the radial distance to the Earth's centre while ϑ and λ are the colatitude and longitude of the observation respectively.

The Swarm satellites follow low Earth orbits (LEO) with altitudes 430–510 km above the Earth's surface, corresponding to radial distances of about $r_{\text{sat}} \approx 1.08a$, where a is the radius of the Earth. They are equipped with vector and scalar magnetometers, which capture high-resolution measurements of the geomagnetic field. It is important to note that the measured magnetic field is not solely generated by the Earth's core, but also has contributions from external sources, particularly the ionosphere and magnetosphere. We will later find that these contributions violate the proposed forward model and require the introduction of robust data weighting in the inversion.

3 Theory — Physics and Forward Model

Before an inverse problem can be solved, we must present the corresponding forward model of the system, which may be formulated in the linear, discretised case as

$$\hat{\boldsymbol{d}} = \boldsymbol{d} + \boldsymbol{e} = \mathrm{G}\boldsymbol{m} + \boldsymbol{e}, \tag{1}$$

where $\mathrm{G} : \mathcal{M} \to \mathcal{D}$ is the *system matrix* which maps model parameters $\boldsymbol{m}$ from the discrete model space, $\mathcal{M} = \mathbb{R}^{N_m}$, to a vector $\boldsymbol{d} = \mathrm{G}\boldsymbol{m}$ in the observation space, $\mathcal{D} = \mathbb{R}^{N_d}$. Here, $\boldsymbol{d}$ is understood to be the *deterministic* component of the forward projection, and is colloquially — and somewhat inaccurately — referred to as the *noise-free data*. Naturally, we do not observe the forward projection of some idealised physical model, but rather observations $\hat{\boldsymbol{d}}$ which also contain contributions from both the realisations of influencing stochastic processes and any systematic biases, both of which are captured by $\boldsymbol{e}$. As we will later discover, $\boldsymbol{e}$ cannot reasonably be assumed to be normally distributed.

In order to determine the system matrix G, we must consider carefully the physical model of the system. To this end, we start from the Ampère-Maxwell law and the assumption that the region of measurement — the orbits upon which the radial component B_r is measured using the *Swarm* satellites — is devoid of free charges and currents. That is, we assert that the magnetic field is curl-free:

$$\nabla \times \boldsymbol{B} = \mu_0 \left(\boldsymbol{J} + \varepsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \right) = \boldsymbol{0}. \tag{2}$$

Thus, we may define a *scalar magnetic potential*, V, such that

$$\boldsymbol{B} = -\nabla V. \tag{3}$$

Deeming it unlikely that any magnetic monopoles will present themselves, we may conclude that the field is also divergence-free, $\nabla \cdot \boldsymbol{B} = 0$, which yields the Laplace equation for the scalar potential:

$$\nabla \cdot \boldsymbol{B} = -\nabla^2 V = 0. \tag{4}$$

If we make the simplifying assumption that a spherical boundary surface of radius a exists which delineates the electromagnetic sources from empty space, we find that the spatial domain of our model is the sphere, S^2. While such a spherical boundary is reasonable for the case of the Earth, assuming that no sources — here understood to be magnetic moments and electrical currents — exist outside of it is more difficult to justify. In particular, this neglects the charge carriers in the ionosphere and the influence of the magnetosphere, a discussion which we will return to later.

With this in mind, we may choose the spherical harmonics $Y_n^m(\vartheta, \lambda)$ as a convenient, orthogonal basis for the model space and continue our analysis in spherical coordinates (r, ϑ, λ), in which we recall Laplace's equation to be of the form:

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \vartheta} \frac{\partial}{\partial \vartheta} \left(\sin \vartheta \frac{\partial V}{\partial \vartheta} \right) + \frac{1}{r^2 \sin^2 \vartheta} \frac{\partial^2 V}{\partial \lambda^2} = 0. \tag{5}$$

Separation of variables and the assumption of interior sources yield the solution [2, p. 507, eq. 13]

$$V(r, \vartheta, \lambda) = a \sum_{n=1}^{\infty} \left(\frac{a}{r} \right)^{n+1} \sum_{m=0}^{n} (g_n^m \cos m\lambda + h_n^m \sin m\lambda) P_n^m(\cos \vartheta) \tag{6}$$

in which P_n^m are the associated Legendre polynomials and we have used the spherical harmonics as given by:

$$Y_{n,\text{even}}^m = \cos(m\lambda) P_n^m(\cos \vartheta) \qquad Y_{n,\text{odd}}^m = \sin(m\lambda) P_n^m(\cos \vartheta). \tag{7}$$

The parameters g_n^m, h_n^m of Equation 6 are referred to as the *Gauss coefficients* and the inverse problem is thus "simply" to determine these given the field measurements.

We do not, however, have access to the potential V directly, which is introduced more as a mathematical convenience than a physical quantity. Instead, we measure the radial component of its gradient as given by Equation 3,

$$B_r(r, \vartheta, \lambda) = -\frac{\partial V}{\partial r} = \sum_{n=1}^{\infty} (n+1) \left(\frac{a}{r} \right)^{n+2} \sum_{m=0}^{n} (g_n^m \cos m\lambda + h_n^m \sin m\lambda) P_n^m(\cos \vartheta). \tag{8}$$

By identifying this as forward projection and truncating the spherical harmonic expansion to degree N, we obtain the semi-discrete model in terms of coordinates:

$$B_r(r,\vartheta,\lambda) = \sum_{n=1}^{N}(n+1)\left(\frac{a}{r}\right)^{n+2}\sum_{m=0}^{n}(g_n^m\cos m\lambda + h_n^m \sin m\lambda)P_n^m(\cos\vartheta), \tag{9}$$

while the finite-dimensional model parameters are consequently given by

$$\boldsymbol{m} = \left(g_1^0, g_1^1, h_1^1, g_2^0, ..., g_N^N, h_N^N\right)^T \in \mathcal{M} \tag{10}$$

where notably the h_n^0 coefficients are omitted as their terms vanish due to the $\sin(m\lambda)$ factor in Equation 6. This truncation is sensible under the assumption that the variations of the magnetic fields are smooth and that the features of interest are of sufficiently low frequency to be captured by the remaining basis functions.

By constructing a system of equations using Equation 9 and the associated discrete model parameters $\boldsymbol{m}$ over an idealised, noise-free set $D = \{d_i = B_r(r_i,\vartheta_i,\lambda_i)\}_{i=1}^{N_d}$, we produce the discrete forward operator as the system matrix, G, which is built as follows:

$$\boldsymbol{d} = \mathrm{G}\boldsymbol{m}$$

$$\begin{pmatrix} B_r(\boldsymbol{x}_1) \\ B_r(\boldsymbol{x}_2) \\ \vdots \\ B_r\left(\boldsymbol{x}_{N_d}\right) \end{pmatrix} = \begin{bmatrix} \Xi_1^0(\boldsymbol{x}_1) & \Xi_1^1(\boldsymbol{x}_1) & \xi_1^1(\boldsymbol{x}_1) & ... & \Xi_N^N(\boldsymbol{x}_1) & \xi_N^N(\boldsymbol{x}_1) \\ \Xi_1^0(\boldsymbol{x}_2) & \Xi_1^1(\boldsymbol{x}_2) & \xi_1^1(\boldsymbol{x}_2) & ... & \Xi_N^N(\boldsymbol{x}_2) & \xi_N^N(\boldsymbol{x}_2) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \Xi_1^0\left(\boldsymbol{x}_{N_d}\right) & \Xi_1^1\left(\boldsymbol{x}_{N_d}\right) & \xi_1^1\left(\boldsymbol{x}_{N_d}\right) & ... & \Xi_N^N\left(\boldsymbol{x}_{N_d}\right) & \xi_N^N\left(\boldsymbol{x}_{N_d}\right) \end{bmatrix} \begin{pmatrix} g_1^0 \\ g_1^1 \\ h_1^1 \\ \vdots \\ g_N^N \\ h_N^N \end{pmatrix} \tag{11}$$

where we have introduced some shorthand notation

$$\begin{aligned} \boldsymbol{x}_i &= (r_i,\vartheta_i,\lambda_i) \\ \Xi_n^m(r,\vartheta,\lambda) &= (n+1)\left(\frac{a}{r}\right)^{n+2}\cos(m\lambda)P_n^m(\cos\vartheta) \\ \xi_n^m(r,\vartheta,\lambda) &= (n+1)\left(\frac{a}{r}\right)^{n+2}\sin(m\lambda)P_n^m(\cos\vartheta). \end{aligned} \tag{12}$$

In conclusion, we can build the discrete forward operator $\mathrm{G} \in \mathbb{R}^{N_d \times N_m}$, $N_m = N(N+2)$ simply by evaluating the basis functions Ξ_n^m and ξ_n^m at the measurement locations, $\{\boldsymbol{x}_i\}_{i=1}^{N_d}$. We note that a final assumption made is that the temporal variations in the field are locally much slower than the period over which the data has been collected, or equivalently that $B_r(r,\vartheta,\lambda)$ is constant with respect to time.

4 Theory — Mathematical Treatment of the Inverse Problem

Having now produced a discrete forward model that may be used with our experimental data, we turn our attention to the associated *inverse problem*, that is, the estimation of model parameters $\boldsymbol{m}$ given the observations $\hat{D} = \left\{\hat{d}_i = B_r(r_i,\vartheta_i,\lambda_i) + e_i\right\}_{i=1}^{N_d}$.

4.1 Least Squares Solution

Naïvely, we may obtain a solution to the inverse problem by the computing the inverse of G with which we would left-multiply to obtain an estimate

$$\widehat{\boldsymbol{m}} = \mathrm{G}^{-1}\hat{\boldsymbol{d}}, \tag{13}$$

except of course such an inversion does not exist since G is not generally square. Instead, we seek the model parameters which minimise the misfit, ϕ, between the observed data $\hat{\boldsymbol{d}}$ and the forward projection in a norm, here taken to be the L_2-norm:

$$\widehat{\boldsymbol{m}} = \arg\min_{\boldsymbol{m}} \phi = \arg\min_{\boldsymbol{m}} \frac{1}{2}\left\|\mathrm{G}\boldsymbol{m} - \hat{\boldsymbol{d}}\right\|_2^2 \tag{14}$$

By inspection, the misfit ϕ is convex, differentiable with respect to $\boldsymbol{m}$, and coercive provided G is full-rank, and thus the (global) minimum may be found by solving

$$\begin{aligned}
\nabla_{\boldsymbol{m}}\phi = 0 &= \nabla_{\boldsymbol{m}}\left[\frac{1}{2}\left(\mathrm{G}\boldsymbol{m} - \hat{\boldsymbol{d}}\right)^{\mathsf{T}}\left(\mathrm{G}\boldsymbol{m} - \hat{\boldsymbol{d}}\right)\right] \\
&= \frac{1}{2}\nabla_{\boldsymbol{m}}\left[-\boldsymbol{m}^{\mathsf{T}}\mathrm{G}^{\mathsf{T}}\hat{\boldsymbol{d}} + \boldsymbol{m}^{\mathsf{T}}\mathrm{G}^{\mathsf{T}}\mathrm{G}\boldsymbol{m} - \hat{\boldsymbol{d}}^{\mathsf{T}}\mathrm{G}\boldsymbol{m} + \hat{\boldsymbol{d}}^{\mathsf{T}}\hat{\boldsymbol{d}}\right] \\
&= \frac{1}{2}\nabla_{\boldsymbol{m}}\left[-2\hat{\boldsymbol{d}}^{\mathsf{T}}\mathrm{G}\boldsymbol{m} + \boldsymbol{m}^{\mathsf{T}}\mathrm{G}^{\mathsf{T}}\mathrm{G}\boldsymbol{m}\right] = -\hat{\boldsymbol{d}}^{\mathsf{T}}\mathrm{G} + \boldsymbol{m}^{\mathsf{T}}\mathrm{G}^{\mathsf{T}}\mathrm{G} \Leftrightarrow \mathrm{G}^{\mathsf{T}}\mathrm{G}\widehat{\boldsymbol{m}} = \mathrm{G}^{\mathsf{T}}\hat{\boldsymbol{d}} \\
\widehat{\boldsymbol{m}} &= \left(\mathrm{G}^{\mathsf{T}}\mathrm{G}\right)^{-1}\mathrm{G}^{\mathsf{T}}\hat{\boldsymbol{d}} = \left(\mathrm{G}^{\mathsf{T}}\mathrm{G}\right)^{-1}\mathrm{G}^{\mathsf{T}}(\boldsymbol{d} + \boldsymbol{e}) = \left(\mathrm{G}^{\mathsf{T}}\mathrm{G}\right)^{-1}\mathrm{G}^{\mathsf{T}}(\mathrm{G}\boldsymbol{m} + \boldsymbol{e}) = \boldsymbol{m} + \left(\mathrm{G}^{\mathsf{T}}\mathrm{G}\right)^{-1}\mathrm{G}^{\mathsf{T}}\boldsymbol{e} \\
&= \boldsymbol{m} + \mathrm{G}^{\dagger}\boldsymbol{e}
\end{aligned} \tag{15}$$

Clearly, our estimate will be rather poor unless $\left\|\mathrm{G}^{\dagger}\boldsymbol{e}\right\| \ll \|\boldsymbol{m}\|$. We can more clearly see how errors may enter our estimate if we consider using the constructed model to compute the radial magnetic field at the Earth's Core-Mantle Boundary (CMB).

4.2 Stability Analysis

Since the model is fully determined by the expansion coefficients in the angular coordinates, this is done simply by constructing a new system matrix $\mathrm{G}_{\mathrm{CMB}}$ as before, but replacing the $\{r_i\}_{i=1}^{N_d}$ radial coordinates with the radius of the CMB, $r_{\mathrm{CMB}} = 3\,480.0\,\mathrm{km}$. By inspection of Equation 11, this will yield an estimate of the field at the boundary, $\hat{\boldsymbol{d}}_{\mathrm{CMB}}$:

$$\hat{\boldsymbol{d}}_{\mathrm{CMB}} = \mathrm{G}_{\mathrm{CMB}}\widehat{\boldsymbol{m}} = \mathrm{G}_{\mathrm{CMB}}\boldsymbol{m} + \mathrm{G}_{\mathrm{CMB}}\mathrm{G}^{\dagger}\boldsymbol{e} = \boldsymbol{d}_{\mathrm{CMB}} + \hat{\boldsymbol{e}} \tag{16}$$

Where $\hat{\boldsymbol{e}} = \mathrm{G}_{\mathrm{CMB}}\mathrm{G}^{\dagger}\boldsymbol{e}$ is the error of our prediction $\hat{\boldsymbol{d}}_{\mathrm{CMB}}$ given the realisation $\boldsymbol{e}$. If we consider a specific spherical degree n and Equation 9, we find that the radial dependence factors out as $(n+1)(a/r)^{n+2}$ which by identity $\mathrm{G}^{\dagger}\mathrm{G} = \mathrm{I}$ implies that $\mathrm{G}^{\dagger}$ must scale by the reciprocal, $(n+1)^{-1}(a/r)^{-n-2}$. If we model all the measurements as having been taken at a single, mean radius, r_{sat}, we thus find

$$\hat{\boldsymbol{e}}_n \propto (n+1)^{-1}\left(\frac{a}{r_{\mathrm{sat}}}\right)^{-n-2}(n+1)\left(\frac{a}{r_{\mathrm{CMB}}}\right)^{n+2}\boldsymbol{e} = \left(\frac{r_{\mathrm{sat}}}{r_{\mathrm{CMB}}}\right)^{n+2}\boldsymbol{e} \tag{17}$$

In our data, the average radius is $r_{\mathrm{sat}} = 6\,891.51\,\mathrm{km}$, giving a spectral noise amplification factor $\eta_n = 1.9803^{n+2}$, $\eta_{20} \approx 3\,370\,000$. Clearly, even minute errors in the high-frequency components of the measurements may dominate our estimate at the core.

This sensitivity to small changes causes the inverse problem to be severely *ill-posed* in the sense of Hadamard and requires us to regularise our inversion.

4.3 Regularisation Theory

A common regularisation scheme is to introduce a penalty on the norm of model parameters, or some linear transformation thereof. In a general case, such an L_p-regularised solution may be found using the variational formulation

$$\arg\min_{\boldsymbol{m}} \frac{1}{2}\left\|\mathrm{G}\boldsymbol{m} - \hat{\boldsymbol{d}}\right\|_{\mathrm{W}}^2 + \frac{\alpha^2}{2}\|\mathrm{H}\boldsymbol{m}\|_p^p, \tag{18}$$

where p is the *norm order*, α^2 is the *regularisation parameter*, and $\mathrm{H} \in \mathbb{R}^{N_k \times N_m}$ is a linear regularisation operator. The norm $\|\boldsymbol{v}\|_{\mathrm{W}}^2 \coloneqq \boldsymbol{v}^{\mathsf{T}}\mathrm{W}\boldsymbol{v}$ indicates the *weighted* L_2 norm where W is a

diagonal matrix of *robust weights* that must be determined jointly in the absence of an *a priori* estimate.

To ensure differentiability at zero, the L_p norm may be approximated using the relaxation scheme due to Ekblom:

$$\|\mathrm{H}\boldsymbol{m}\|_p^p \approx \sum_{i=1}^{N_k} \left[\left((\mathrm{H}\boldsymbol{m})_i \right)^2 + \delta^2 \right]^{p/2}, \tag{19}$$

which yields an approximate solution to Equation 18 of the form [3, eq. 51]

$$\widehat{\boldsymbol{m}}_{r\alpha p} = \left(\mathrm{G}^\mathsf{T}\mathrm{W}\mathrm{G} + \alpha^2 \mathrm{H}^\mathsf{T}\mathrm{W}_{mp}\mathrm{H} \right)^{-1} \mathrm{G}^\mathsf{T}\mathrm{W}\hat{\boldsymbol{d}} = \tilde{\mathrm{G}}_{\alpha p}^{-1}\mathrm{G}^\mathsf{T}\mathrm{W}\hat{\boldsymbol{d}} = \mathrm{G}_{\alpha p}^{-1}\hat{\boldsymbol{d}}. \tag{20}$$

Here, both the robust data weights W and the (diagonal) regularisation weights may be obtained through the Iteratively Reweighted Least Squares (IRLS) algorithm, as described by Algorithm 1. Specifically, the diagonal of W_{mp} is populated by the vector $\boldsymbol{w}_m$, which is updated element-wise at each iteration by

$$\boldsymbol{w}_m = p(\boldsymbol{\gamma}^2 + \delta^2)^{p/2-1}, \quad \text{where} \quad \boldsymbol{\gamma} = \mathrm{H}\boldsymbol{m}, \tag{21}$$

with δ denoting the *Ekblom perturbation* which must satisfy $\delta^2 \ll \|\boldsymbol{\gamma}\|_2^2 / N_k$.

4.4 Robust Iteratively Reweighted Least Squares Algorithm

We propose a version of the standard Iteratively Reweighted Least Squares algorithm which features a number of additional considerations. While we will only use a single regularisation term in this report, the proposal in Algorithm 1 allows an arbitrary number k of regularisation terms to be employed. It also allows for different M-estimators and scale estimators to be used for the computation of the robust weights and the dispersion scale $\hat{\sigma}$ respectively.

For the purposes of this report, we choose the Huber weights [3, p.9] and normalised Median Absolute Deviation (MAD) [4, p.36] as the M-estimator and scale estimator respectively, which are given by $\mathcal{S}(\boldsymbol{\varepsilon}; \boldsymbol{\theta}_\mathcal{S})$ and $\mathcal{V}(\boldsymbol{\varepsilon}; \boldsymbol{\theta}_\mathcal{V})$ in Algorithm 1:

$$w_{\text{Huber}}(\varepsilon; c = 1.345) = \begin{cases} c/|\varepsilon| & |\varepsilon| > c \\ 1 & |\varepsilon| \leq c \end{cases} \tag{22}$$

$$\hat{\sigma}_{\text{MAD}}(\boldsymbol{\varepsilon}) = \frac{\text{median}\left(\sum_{i=1}^{N_d} |\varepsilon_i - \text{median}(\boldsymbol{\varepsilon})| \right)}{\Phi^{-1}(0.75)} \tag{23}$$

where $\Phi(p)$ is the CDF of a standard Gaussian distribution such that $\Phi^{-1}(0.75) \approx 0.675$ [4, p.36].

As a termination criterion for the algorithm, the relative size of the model parameter update in the L_2-norm is compared against the given tolerance τ at each time step to enable stopping before the iteration limit N_{iter}.

The implementation of Algorithm 1 is provided in the attached code and may also be found in Appendix A.3. This features some additional parameters pertaining to debugging and observability into the performance of the algorithm.

Algorithm 1: Robust Iteratively Reweighted Least Squares (IRLS-R)

procedure IRLS-R(G, $\boldsymbol{d}$, H_k, α_k, δ_k, p_k, $\mathcal{V}$, $\mathcal{S}$, N_{iter}, τ)
  input Model $\mathrm{G} \in \mathbb{R}^{N_m \times N_d}$, data $\boldsymbol{d} \in \mathbb{R}^{N_d}$,
    Regularisers $\mathrm{H}_{k\in\mathcal{K}} \in \mathbb{R}^{N_k \times N_m}$, reg. parameters $\alpha_{k\in\mathcal{K}}$, norm orders $p_{k\in\mathcal{K}}$,
    Ekblom perturbations $\delta_{k\in\mathcal{K}}$,
    M-estimator $\mathcal{S}(\boldsymbol{\varepsilon};\boldsymbol{\theta}_{\mathcal{S}})$, scale estimator $\mathcal{V}(\boldsymbol{\varepsilon};\boldsymbol{\theta}_{\mathcal{V}})$,
    Maximal iterations N_{iter}, tolerance τ
  ▷ $\mathcal{K} = \{1, ..., K\} \subset \mathbb{N}$: regularisation term indices
  output Model parameters $\boldsymbol{m} \in \mathbb{R}^{N_m}$,
    Robust weights $\boldsymbol{w} \in \mathbb{R}^{N_d}$, Ekblom measures $\boldsymbol{\gamma}_{k\in\mathcal{K}}$
  ▷ Initialisation
  $\boldsymbol{m} \leftarrow \mathbb{0}_{N_m}$
  $\boldsymbol{w} \leftarrow \mathbb{1}_{N_d}$
  $\boldsymbol{\gamma}_k \leftarrow \mathbb{0}_{N_k} \quad \forall k \in \mathcal{K}$
  for $i = 1$ to N_{iter} **do**
    ▷ Construct system matrix and right-hand side, $\mathrm{A}\boldsymbol{m}_{\text{new}} = \boldsymbol{b}$
    $\boldsymbol{w}_{mk} \leftarrow p_k(\boldsymbol{\gamma}_k^2 + \delta_k^2)^{p_k/2-1} \quad \forall k$ ▷ Compute Ekblom norm weights
    $\mathrm{A} \leftarrow \mathrm{G}^{\mathsf{T}} \operatorname{diag}(\boldsymbol{w})\mathrm{G} + \sum_{k=1}^{K} \alpha_k^2 \mathrm{H}_k^{\mathsf{T}} \operatorname{diag}(\boldsymbol{w}_{mk})\mathrm{H}_k$ ▷ System matrix, A
    $\boldsymbol{b} \leftarrow \mathrm{G}^{\mathsf{T}}(\boldsymbol{w} \odot \boldsymbol{d})$ ▷ RHS, $\boldsymbol{b}$; $\odot$ is element-wise mult.
    $\boldsymbol{m}_{\text{new}} \leftarrow \textsc{Solve}(\mathrm{A}, \boldsymbol{b})$ ▷ $\mathrm{A} \succ 0$, $\mathrm{A}^{\mathsf{H}} = \mathrm{A}$, $\therefore$ use Cholesky
    $\Delta_2 \leftarrow \|\boldsymbol{m} - \boldsymbol{m}_{\text{new}}\|_2 / \|\boldsymbol{m}\|_2$ ▷ Convergence criterion
    ▷ Update model
    $\boldsymbol{m} \leftarrow \boldsymbol{m}_{\text{new}}$
    ▷ Update weights
    $\boldsymbol{\varepsilon} \leftarrow \boldsymbol{d} - \mathbf{G}\boldsymbol{m}$ ▷ Compute unweighted residual
    $\hat{\sigma} \leftarrow \mathcal{V}(\boldsymbol{\varepsilon})$ ▷ Estimate data dispersion scale
    $\boldsymbol{w} \leftarrow \mathcal{S}(\boldsymbol{\varepsilon}/\hat{\sigma})$ ▷ Compute robust weights
    $\boldsymbol{\gamma}_k \leftarrow \mathrm{H}_k\boldsymbol{m} \quad \forall k$ ▷ Compute Ekblom measures
    if $\Delta_2 < \tau$ **then** ▷ Stop if converged
      BREAK
    end
  end
  return $\boldsymbol{m}, \boldsymbol{w}, \boldsymbol{\gamma}_{k\in\mathcal{K}}$
end

4.5 Covariance and Resolution Matrices

We wish to be able to compute estimates of the covariance and resolution matrices of our model, which will give insight into respectively the uncertainty and bias of our obtained model, $\widehat{\boldsymbol{m}}$. Additionally, we may propagate the model covariance onto the prediction to produce estimates of the uncertainty of the obtained radial magnetic field.

We first note that $\tilde{\mathrm{G}}_{\alpha p}^{-1} = \left(\mathrm{G}^{\mathsf{T}}\mathrm{W}\mathrm{G} + \alpha^2 \mathrm{H}^{\mathsf{T}}\mathrm{W}_{mp}\mathrm{H}\right)^{-1}$ is symmetric by the diagonality of $\mathrm{W}, \mathrm{W}_{mp}$. Thus, by identity $\operatorname{Cov}(\mathrm{A}\boldsymbol{b}) = \mathrm{A}\operatorname{Cov}(\boldsymbol{b})\mathrm{A}^{\mathsf{T}}$, we obtain an estimated model covariance matrix

$$\begin{aligned}
\Sigma_{\widehat{\boldsymbol{m}}} = \operatorname{Cov}(\widehat{\boldsymbol{m}}_{r\alpha p}) &\approx \mathrm{G}_{\alpha p}^{-1} \operatorname{Cov}\left(\hat{\boldsymbol{d}}\right)\left(\mathrm{G}_{\alpha p}^{-1}\right)^{\mathsf{T}} \\
&= \tilde{\mathrm{G}}_{\alpha p}^{-1}\mathrm{G}^{\mathsf{T}}\mathrm{W}\operatorname{Cov}\left(\hat{\boldsymbol{d}}\right)\left(\tilde{\mathrm{G}}_{\alpha p}^{-1}\mathrm{G}^{\mathsf{T}}\mathrm{W}\right)^{\mathsf{T}} \\
&= \tilde{\mathrm{G}}_{\alpha p}^{-1}\mathrm{G}^{\mathsf{T}}\mathrm{W}\operatorname{Cov}\left(\hat{\boldsymbol{d}}\right)\mathrm{W}\mathrm{G}\tilde{\mathrm{G}}_{\alpha p}^{-1}.
\end{aligned} \tag{24}$$

Here W denotes the *frozen* robust weights obtained at the final iteration of Algorithm 1, which provide a reasonable, linear approximation in this case.

The variance of the measurements are not known a priori and we resort to a *plug-in* estimate in which we assume the data to be uncorrelated and homoscedastic with a variance approximated by the Median Absolute Deviation of the residuals, which has been jointly estimated as part of Algorithm 1:

$$\mathrm{Cov}\big(\hat{\boldsymbol{d}}\big) \approx \Big[\hat{\sigma}_{\mathrm{MAD}}\big(\hat{\boldsymbol{d}} - \mathrm{G}\widehat{\boldsymbol{m}}\big)\Big]^2 \cdot \mathrm{I}_{\mathrm{N_d}}. \tag{25}$$

Similarly, by applying the treatment used to obtain [3, eq. 35], the resolution matrix, $\mathrm{R}_{\widehat{\boldsymbol{m}}}$, may be computed using

$$\mathrm{R}_{\widehat{\boldsymbol{m}}} \approx \mathrm{G}_{\alpha p}^{-1}\mathrm{G} = \tilde{\mathrm{G}}_{\alpha p}^{-1}\mathrm{G}^{\mathsf{T}}\mathrm{W}\mathrm{G} \tag{26}$$

where we recall that the limiting case $\mathrm{R}_{\widehat{\boldsymbol{m}}} = \mathrm{I}$ identifies the optimal case where the data is fully resolved by the inversion.

Finally, we may estimate the covariance matrix of our reconstruction at the Core-Mantle Boundary, $\hat{\boldsymbol{d}}_{\mathrm{CMB}} = \mathrm{G}_{\mathrm{CMB}}\widehat{\boldsymbol{m}}$ as

$$\begin{aligned}
\Sigma_{\hat{\boldsymbol{d}}_{\mathrm{CMB}}} &\approx \mathrm{G}_{\mathrm{CMB}}\Sigma_{\widehat{\boldsymbol{m}}}\mathrm{G}_{\mathrm{CMB}}^{\mathsf{T}} \\
&= \mathrm{G}_{\mathrm{CMB}}\tilde{\mathrm{G}}_{\alpha p}^{-1}\mathrm{G}^{\mathsf{T}}\mathrm{W}\,\mathrm{Cov}\big(\hat{\boldsymbol{d}}\big)\mathrm{W}\mathrm{G}\tilde{\mathrm{G}}_{\alpha p}^{-1}\mathrm{G}_{\mathrm{CMB}}^{\mathsf{T}} \\
&= \Big[\hat{\sigma}_{\mathrm{MAD}}\big(\hat{\boldsymbol{d}} - \mathrm{G}\widehat{\boldsymbol{m}}\big)\Big]^2\mathrm{G}_{\mathrm{CMB}}\tilde{\mathrm{G}}_{\alpha p}^{-1}\mathrm{G}^{\mathsf{T}}\mathrm{W}^2\mathrm{G}\tilde{\mathrm{G}}_{\alpha p}^{-1}\mathrm{G}_{\mathrm{CMB}}^{\mathsf{T}}.
\end{aligned} \tag{27}$$

where the scalar data covariance and diagonal weight matrix are exploited in the last step to significantly reduce the computational burden.

5 Results

We will investigate the performance of three different inversions starting out with the unregularised least-squares solution, which can be expected to perform poorly due to the illposedness of the problem. To promote stability in the inversion, we subsequently present and analyse the results of both regularisation in both the L_2 and L_1 norms.

5.1 Unregularised Least Squares Solution

If we disregard the analysis carried out in Section 4 and attempt to construct a model by inversion using the naïve, unregularised least-squares solution given by Equation 15, we recover the reconstruction of the radial magnetic field at the CMB shown in Figure 2.

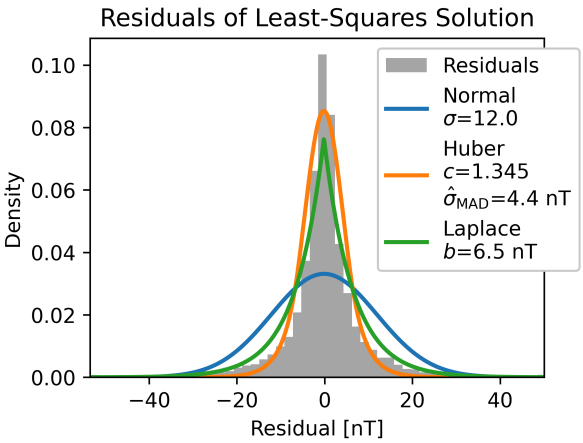


Figure 1: Residual of naïve least squares solution.

Immediately, we note the rapid oscillations exhibited by the field towards the poles, which does not agree with our expectations. Further, the magnitude of the field oscillations is much larger than that found in the literature [5], which prescribes an amplitude of approximately 1 mT. This is a consequence of the aforementioned ill-posedness of the inverse problem. Further, inspection of the residuals $\boldsymbol{\varepsilon} = \hat{\boldsymbol{d}} - \mathrm{G}\widehat{\boldsymbol{m}}$ as shown in Figure 1 reveals that these are far from normally distributed and instead appear to follow a long-tailed distribution such as the Huber distribution, which has been fitted to the data alongside the Laplace distribution.

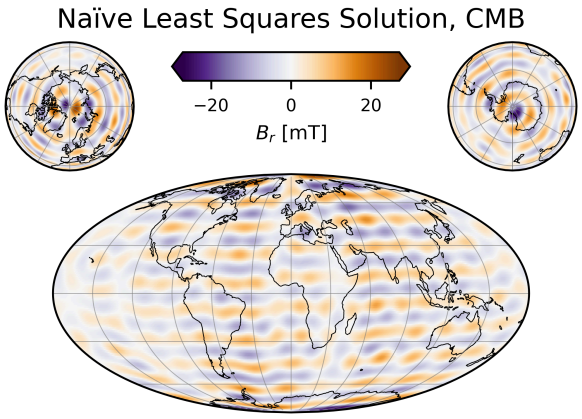


Figure 2: Bad reconstruction of radial magnetic field at CMB due to unstable inversion.

It is for this reason that we have chosen the Huber weights given in Equation 22 as the M-estimator for our IRLS algorithm, which has the effect of reducing the influence of observations further from the centre of the data distribution.

5.2 Non-Robust L_2-Regularised Solution

We consider the specific case of a non-robust L_2-regularised solution, which by the treatment in Section 4 and identifying

$$p = 2, \qquad \mathrm{W} = \mathrm{W}_{\mathrm{m2}} = \mathrm{I} \tag{28}$$

gives

$$\begin{aligned} \boldsymbol{m}_{\alpha 2} &= \left(\mathrm{G}^{\mathsf{T}}\mathrm{W}\mathrm{G} + \alpha^2 \mathrm{H}^{\mathsf{T}}\mathrm{W}_{\mathrm{m2}}\mathrm{H}\right)^{-1}\mathrm{G}^{\mathsf{T}}\mathrm{W}\boldsymbol{d} \\ &= \left(\mathrm{G}^{\mathsf{T}}\mathrm{G} + \alpha^2 \mathrm{H}^{\mathsf{T}}\mathrm{H}\right)^{-1}\mathrm{G}^{\mathsf{T}}\boldsymbol{d} = \tilde{\mathrm{G}}_{\alpha 2}^{-1}\mathrm{G}^{\mathsf{T}}\boldsymbol{d} = \mathrm{G}_{\alpha 2}^{-1}\boldsymbol{d} \end{aligned} \tag{29}$$

where $\mathrm{G}_{\alpha 2}^{-1} = \left(\mathrm{G}^{\mathsf{T}}\mathrm{G} + \alpha^2 \mathrm{H}^{\mathsf{T}}\mathrm{H}\right)^{-1}\mathrm{G}^{\mathsf{T}}$ may be used to find the covariance and resolution matrices through Equation 24 and Equation 26 respectively. Notably, since neither of the iteratively constructed weight matrices feature, this regularization can be carried out without resorting to Algorithm 1.

To select an appropriate regularization parameter, the L-curve method is applied. Figure 3 shows the L-curve as a scatter plot, where the corner identifies the optimal trade-off between data misfit and model norm. The corresponding regularization parameter is

$$\alpha^2 = 1.60 \times 10^{-8}. \tag{30}$$

The non-robust L_2-regularized solution is obtained from the closed-form expression derived above, which arises from minimizing a quadratic objective function.

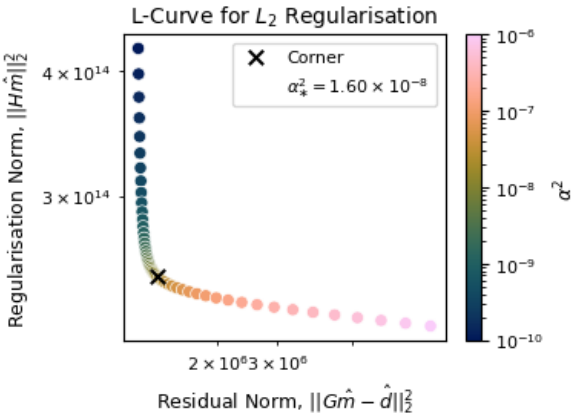
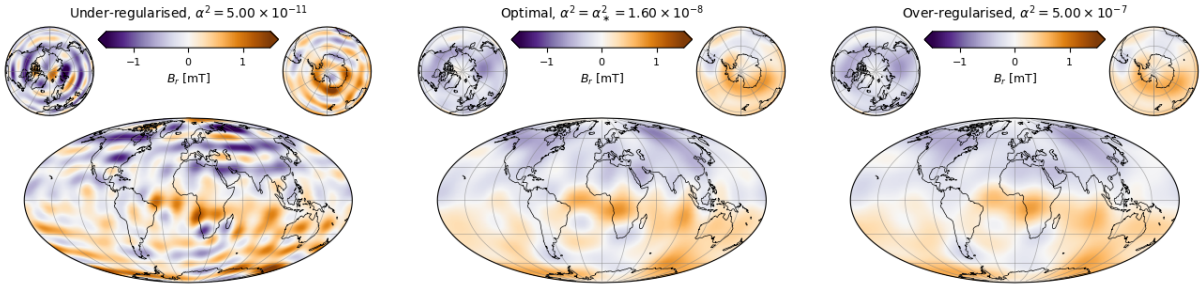


Figure 3: L-curve for the L_2 least-squares solution

This value provides a reasonable balance between fitting the observations and suppressing instability that would otherwise produce dominant and noisy high-frequency spatial oscillations.



(a) Under-regularized, $\alpha^2 = 5 \times 10^{-11}$.

(b) Optimal regularized, $\alpha_*^2 = 1.60 \times 10^{-8}$

(c) Over-regularized, $\alpha^2 = 5 \times 10^{-7}$

Figure 4: Solutions to the non-robust L_2-regularized problem for various regularization parameters.

To illustrate the effect of regularization, three representative solutions are considered: an under-regularized solution, the optimal solution, and an over-regularized solution, shown in Figure 4. For small values of α^2, the regularization is too weak and the inversion fits the data too closely. This leads to overly oscillatory behavior and amplification of high-degree spherical harmonic components, producing unrealistic small-scale features in the CMB field. For large values of α^2, the regularization becomes too strong and suppresses regions with high spatial frequencies model parameters excessively, resulting in an overly smooth field. The optimal solution, as shown in Figure 5, lies between these two extremes and notably preserves the large reversed-flux region over the South Atlantic Ocean while reducing noise.

Figure 6 shows the power spectra for different values of α^2. When α^2 is small, the spectrum retains relatively strong high-degree contributions, indicating that high-frequency structure and noise are insufficiently damped. When α^2 is large, the high-degree components are strongly suppressed and the spectrum is dominated by low-degree terms.

Among the examined solutions, the case with $\alpha_*^2 = 1.60 \times 10^{-8}$ shown in green provides a good balance between retaining the structure of the reconstruction while suppressing high-frequency spatial fluctuations, as also reflected by its intermediate decay behavior in the power spectrum.

The residual distribution for the optimal L_2-regularized solution is shown in Figure 7. It is strongly peaked around zero and approximately symmetric, but is clearly not normally distributed. However, the residuals exhibit heavier tails than a Gaussian distribution and are more closely matched by a Laplace distribution. This suggests that the errors are not purely Gaussian and may include modelling errors, outliers, or unresolved physical processes. This behavior highlights a limitation of the non-robust L_2 formulation and motivates the use of robust L_1 regularization.

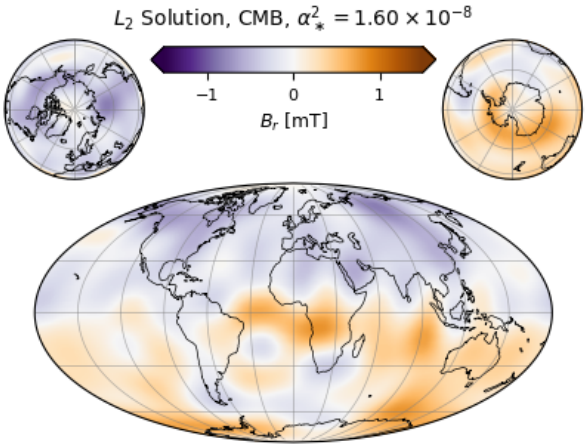


Figure 5: Projection of L_2-regularised solution onto CMB

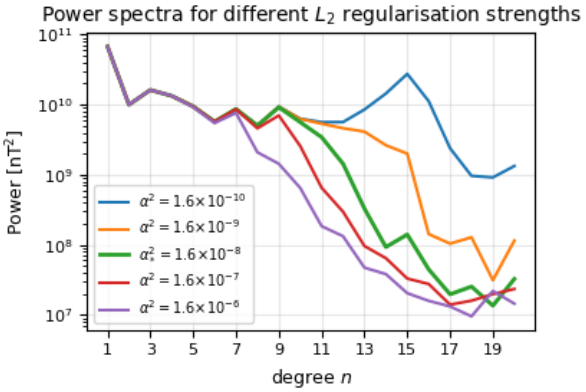


Figure 6: Power spectra for different L_2 regularization strengths

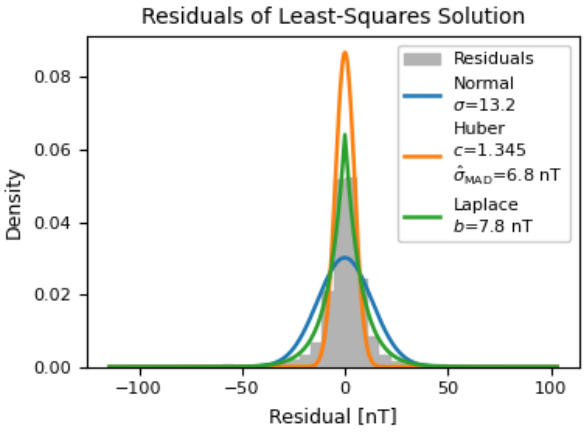


Figure 7: Distribution of residuals for the L_2-regularised model

5.3 Robust L_1-Regularised Solution

For the L_1-regularised solution with robust weights, we require the full machinery described in Section 4.3 and must resort to iteratively approaching the correct weights $\mathrm{W}, \mathrm{W}_{\mathrm{m1}}$ using the IRLS algorithm given in Algorithm 1 with the norm order given by $p = 1$.

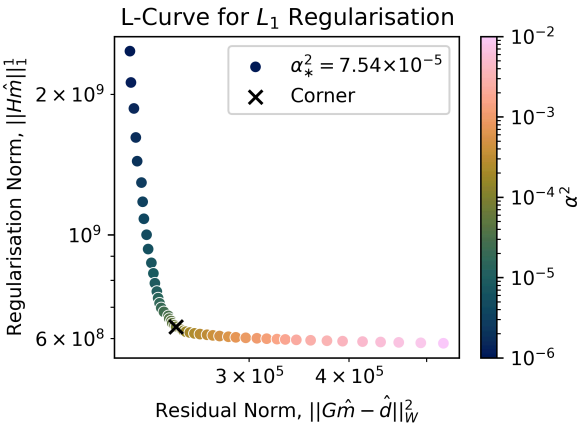
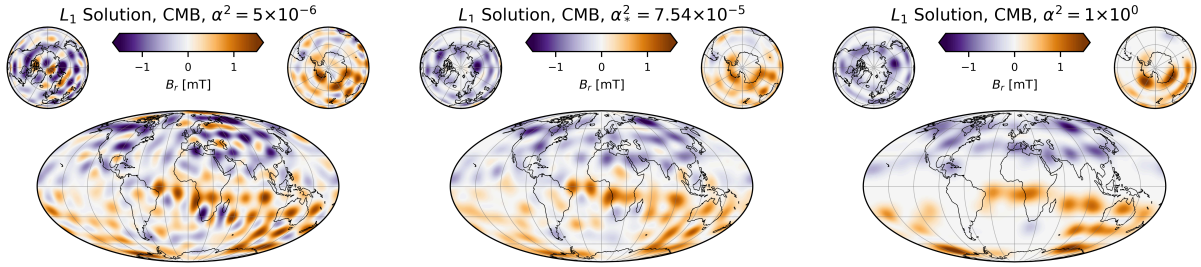


Figure 8: L-curve for the robust L_1-regularised solutions. The corner is determined using the orthogonal chord distance.

We again seek to find an optimal regularisation parameter α^2 by employing the L-curve heuristic, which must notably be carried out with the appropriately weighted norms. That is, we must use $\left\|\mathrm{G}\widehat{\boldsymbol{m}}_{r\alpha 1} - \hat{\boldsymbol{d}}\right\|_{\mathrm{W}}^2$ and $\|\mathrm{H}\widehat{\boldsymbol{m}}\|_1$ to produce the L-curve on which we seek the corner by finding the point with the largest orthogonal distance to the chord between the extremal values of α. This method offers a more robust method of estimating the L-curve corner than those seeking the maximal curvature, which were observed to be inaccurate in some cases due to their reliance on estimating the second derivative.

The regularisation parameter with optimal trade-off between the regularisation and data fidelity norms is found to be $\alpha_*^2 = 7.54 \times 10^{-5}$ and gives the solution seen in Figure 9 alongside underregularised and overregularised solutions, where Figure 9a and Figure 9b are computed to tolerance $\tau = 10^{-9}$ and Figure 9c is computed for $\tau = 5 \times 10^{-7}$.

We note that insufficient regularisation pushes the solution towards the oscillatory, unstable regime first seen in Section 5.1, whereas the over-regularised solution promotes sparsity to the extent that important features are removed. A larger plot of the optimal solution is available in Appendix A.2. For the remainder of the analysis, we select the parameter $\alpha_*^2 = 7.54 \times 10^{-5}$.



(a) Under-regularised, $\alpha^2 = 5 \times 10^{-6}$

(b) Optimal regularisation, $\alpha_*^2 = 7.54 \times 10^{-5}$

(c) Over-regularised, $\alpha^2 = 1$

Figure 9: Solutions to the robust L_1-regularised problem for various regularisations parameters.

If we inspect the power spectrum of the solution shown in Figure 10, we find that the contribution to the total power spectrum decreases for higher-order components of the model — this is encouraging, as a diverging power spectrum would indicate that the truncated orders are significant and the model derived in Section 3 would need to be reconsidered in this case. The fact that most of the power is contained in the first degree can be understood through the largely dipolar shape of the obtained field, where

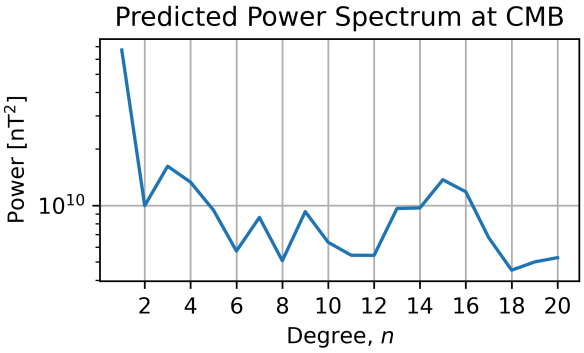


Figure 10: Power spectrum of the robust L_1-regularised solution. Note in particular that the power decreases for higher degrees.

the field lines mostly leave the core in the southern hemisphere and enter it in the northern hemisphere. A perfect dipole would, of course, be captured wholly by the first degree, as evident by Equation 9.

It is also worthwhile to consider the rate of decay in the components of Figure 10, which has largely leveled off at a high constant value of $\approx 4 \cdot 10^9\ \mathrm{nT}^2$ at higher degrees. This is somewhat concerning as it implies that the truncated degrees are not entirely insignificant to the prediction of the field at the CMB. We will contemplate this further below and return to this point in the discussion.

5.3.a Covariance and Resolution Characteristics

When also considering higher degrees, we recover a region of flux reversed relative to the simple dipole model that may be considered as a null hypothesis. In order to give credence to its existence, we first compute an estimate of model covariance matrix, $\Sigma_{\widehat{\boldsymbol{m}}}$ by the method described in Section 4.5. We note again that the estimate is produced under a number of assumptions about the data, in particular that the data errors are uncorrelated and have a common variance, which may not be the case if the dominant source of errors are local influences from the ionosphere and magnetosphere.

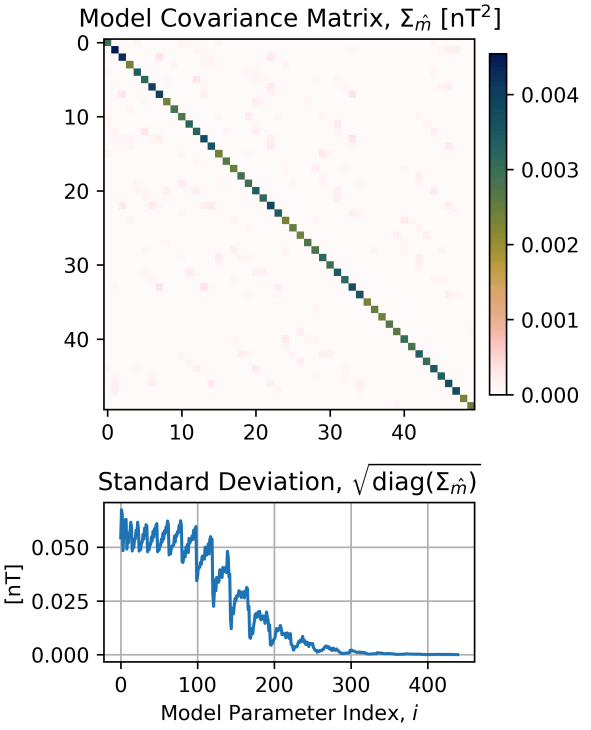


Figure 11: The model covariance matrix and its associated standard deviations. Saw-tooth pattern arises from increments in degree of spherical harmonics.

The first 50 indices of the model covariance are shown in Figure 11 where the diagonal elements can be seen to dominate until approximately the 14th degree. This is at least in part a result of having used the *plug-in* estimate for the data covariance $\Sigma_{\boldsymbol{d}}$ as forewarned. In the lower figure the ordering of the Gauss coefficients becomes apparent by the jagged nature of the parameter standard deviations. The flattened parameter index i follows the natural ordering of Equation 9 and each saw-tooth of the plot is attributed to an increment of the degree $0 < n \leq N$ with the associated orders $0 \leq m \leq n$ immediately following it. The mapping from the first Gauss coefficients of each degree, g_n^0, and the model parameter index i can be found in Table 1. In conclusion, higher indices generally belong to higher frequency components of the model.

g_n^0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
i	0	3	8	15	24	35	48	63	80	99	120	143	168	195	224	255	288	323	360	399

Table 1: Mapping from the first Gauss coefficient of each degree n to the model parameter index i.

It may be initially surprising that the standard deviation decreases for higher-frequency Gauss coefficients, but this arises from the bias introduced by the regularisation. As the harmonic degree n increases, the signal at the satellite becomes increasingly attenuated by the $(a/r)^{n+2}$ factor of Equation 9 since $a/r_{\mathrm{sat}} \approx 0.925$. This is what leads to instability in the unregularised problem as discussed in Section 4.2, but in the regularised problem these terms are instead dominated by the regularisation scheme, which penalises the size of the model parameters in the L_1 norm and thus promotes the sparsity that is also reflected directly in the projection onto the CMB. This drives the high-frequency coefficients towards zero and explains the phenomenon observed in Figure 11 and gives a theoretical understanding to the over-regularised case shown in Figure 9c.

To give credence to this hypothesis, we compute the model resolution matrix, $\mathrm{R}_{\widehat{\boldsymbol{m}}}$, which is depicted in Figure 12. It largely follows the same shape as Figure 11, but the interpretation is different — where the covariance matrix contains information about the variance of the model parameters, the resolution matrix offers insights into the *bias* of the model. This can be understood by its construction from the pseudo-inverse and forward projection, which probes the null-space of the inversion. Diagonal entries close to unity correspond with perfectly resolved parameters, whereas values close to zero indicate that only very little information about those parts of the model are recovered. Off-diagonal elements are understood to capture the model's ability to discern how information of the measurement data is distributed onto the parameters, where higher values indicate a larger influence of one parameter on the other.

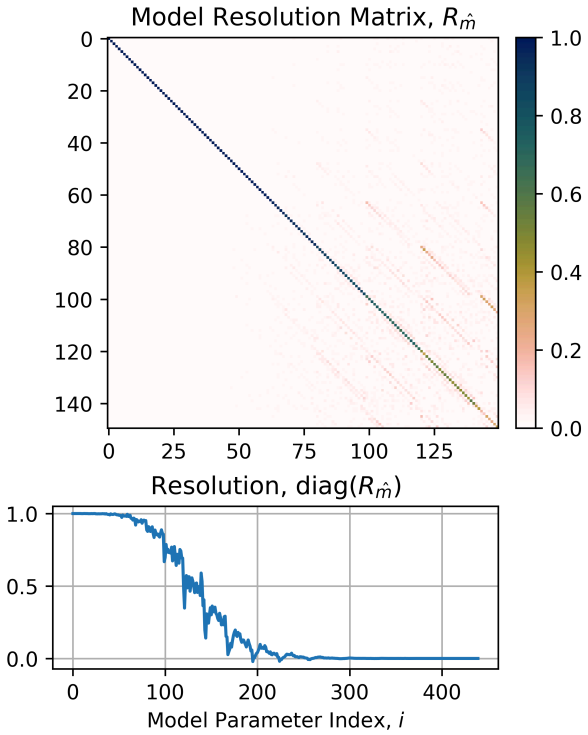
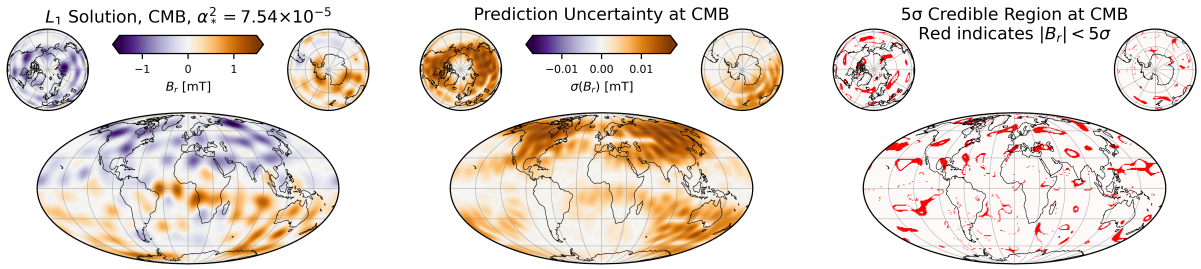


Figure 12: Model resolution matrix and its diagonal. Jagged edges in diagonal arise from increments in the degree of the spherical harmonics.

We observe that higher frequency components have lower resolutions and are almost entirely lost during the inversion as a result of the strong attenuation discussed in Section 4.2. This also suggests that even if higher degrees were included in the model, we would not be able to derive significant information from them given the same dataset, $\hat{D}$ — the higher degrees are already dominated by the bias introduced through the regulariser.

In conclusion, we find that the model has poor resolution of smaller spatial features and that the reconstruction of those are dominated by the a priori choice of regulariser.

Lastly, we may compute estimates of the variance in the magnetic field predictions at the CMB, which are the diagonal entries of $\Sigma_{\hat{\boldsymbol{d}}_{\mathrm{CMB}}}$ described in Section 4.5. From these we can produce a map over the estimated standard errors of our prediction, as shown in Figure 13b, where we note that the region with reversed flux has standard errors lower than 0.1 mT. From this, we can produce the binary mask shown in Figure 13c, which demarcates regions of higher and lower confidence, here given by the conservative threshold that the prediction must lie further than 5 standard deviations from zero to be considered credible.



(a) The L_1 regularised solution

(b) Estimated standard error of predicted field at CMB

(c) Regions where prediction is larger than 5 standard deviations

Figure 13: Investigation into credibility of L_1-regularised solution.

We find that our prediction satisfies this threshold for most of the surface of the CMB, but struggles to confidently recover the high-frequency edges of some patches, which agrees with the previous analysis of the model covariance and resolution matrices. In particular, we note that the region of re-

versed flux beneath the southern Atlantic satisfies the confidence threshold, indicating that the observed phenomenon is not merely an artefact of the inversion.

6 Discussion

Comparing the predicted field at the CMB produced by the L_2 and robust L_1 model, best viewed in Appendix A.1 and Appendix A.2 respectively, we clearly observe the effect of the different regularisations schemes for the high-frequency components, which we found to have high variance and low resolution in Section 5.3. Where the L_2-regularisation steers the solution towards a smoothly varying surface, the L_1-regulariser promotes sparsity in the model instead, which manifests as regions with only minute radial magnetic fields.

However, for those parts of the model that are well-represented in the data, we find remarkable agreement between the two models. In particular, we note that the region of reversed flux in the southern Atlantic is found in both models, which further supports a rejection of the dipole model taken as the null-hypothesis. The region is large, and is thus not likely to be a feature of the regularisation bias, given the relatively low spatial frequencies associated with it. By the analysis in Section 5.3, we can thus expect the region to be well-resolved by at least the model regularised in the L_1-norm.

The analysis of the prediction error carried out in Section 5.3 provides very high confidence in the existence of a region of reversed flux given the 5σ confidence threshold map of Figure 13c, though as noted this result relies on both a plug-in estimate obtained from the Median Absolute Deviation of the residuals in the final iteration of Algorithm 1 as well as the assumption that the data errors are both univariate and uncorrelated. Given our understanding that these errors are likely the result of charges and currents in the ionosphere and magnetosphere — which also violate the interior sources assumption made in Section 3 — it seems unreasonable that they would share a common variance and be entirely uncorrelated. For this reason, it would be unwise to rely too heavily on the obtained estimate of the prediction error, hence the conservative 5σ threshold value.

Additionally, we may return to the concerning finding in Figure 10 that the power carried by the truncated model parameters may be significant. Thankfully, those concerns are largely alleviated by the subsequent analysis regarding the regularisation bias presented in Section 5.3.a, which explains that the Gauss coefficients associated with harmonic degrees larger than approximately $n = 14$ are determined largely by the a priori choice of regulariser. This was also discussed in Section 4.2 and arises due to the attenuation of high-frequency information in the model. Consequently, the truncation of the model does not noticeably diminish the amount of credible information about the geomagnetic behaviour at the CMB made available by our method.

Lastly, we propose that the model may be significantly improved given a larger dataset, or by relaxing the assumption of interior sources in the forward projection to produce a more complex model that can directly incorporate dynamics or data from the ionosphere and magnetosphere. The latter would reduce the model residuals considerably and could be carried out either as a separate contribution to the field or by abandoning the interior sources assumption.

7 Conclusion

We have employed an Iteratively Reweighted Least Squares (IRLS) algorithm with robust reweighting and Ekblom relaxation of the L_1-norm to solve the regularised inverse problem of obtaining the Gauss coefficients of the radial magnetic field expansion on the truncated spherical harmonics of degree $N = 20$ given the assumption of interior sources. This model was constructed using data gathered by the ESA Swarm mission in addition to a similar model using L_2-regularisation without data reweighting.

While the inversion is found to be predominantly determined by the a priori choice of regularisation for components of high spatial frequency, both models confidently produce a region of reversed flux beneath the southern Atlantic when evaluated at the Core-Mantle Boundary (CMB). The prediction error of the L_1-regularised model was studied under the assumption of univariate and uncorrelated data errors, and the estimated field was determined to be above a 5σ threshold over the entire surface with the exception of higher-frequency edges in the reconstruction.

Having considered the influence of the unmodelled ionosphere and magnetosphere, we reject the null-hypothesis of a purely dipolar geomagnetic model and find evidence supporting the existence of the Southern Atlantic Anomaly. We do so with reference to the obtained confidence of prediction and the ability of the proposed model to resolve features of spatial frequencies similar to that of the flux reversal region.

8 Bibliography

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A Appendix

A.1 Large Plot of Solution to L_2-regularised Problem

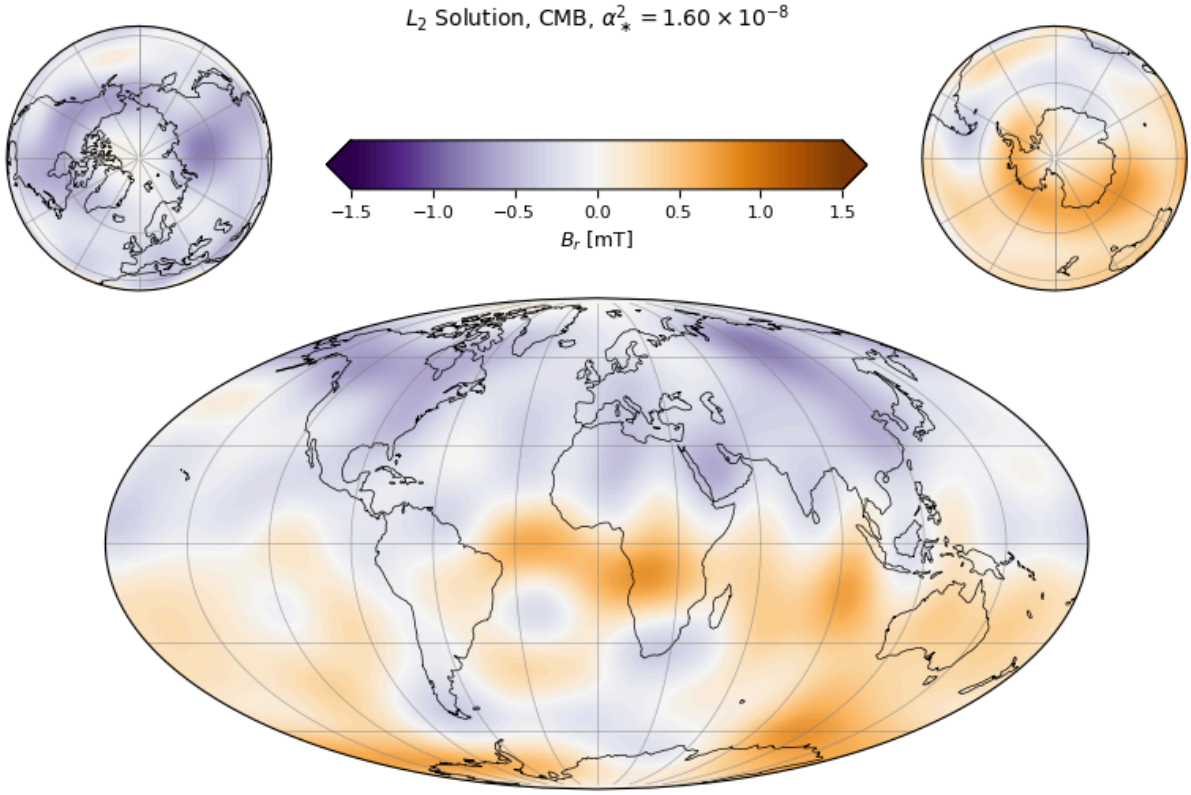


Figure 14: Large version of the plot shown in Figure 4 (b)

A.2 Large Plot of Solution to Robust L_1-regularised Problem

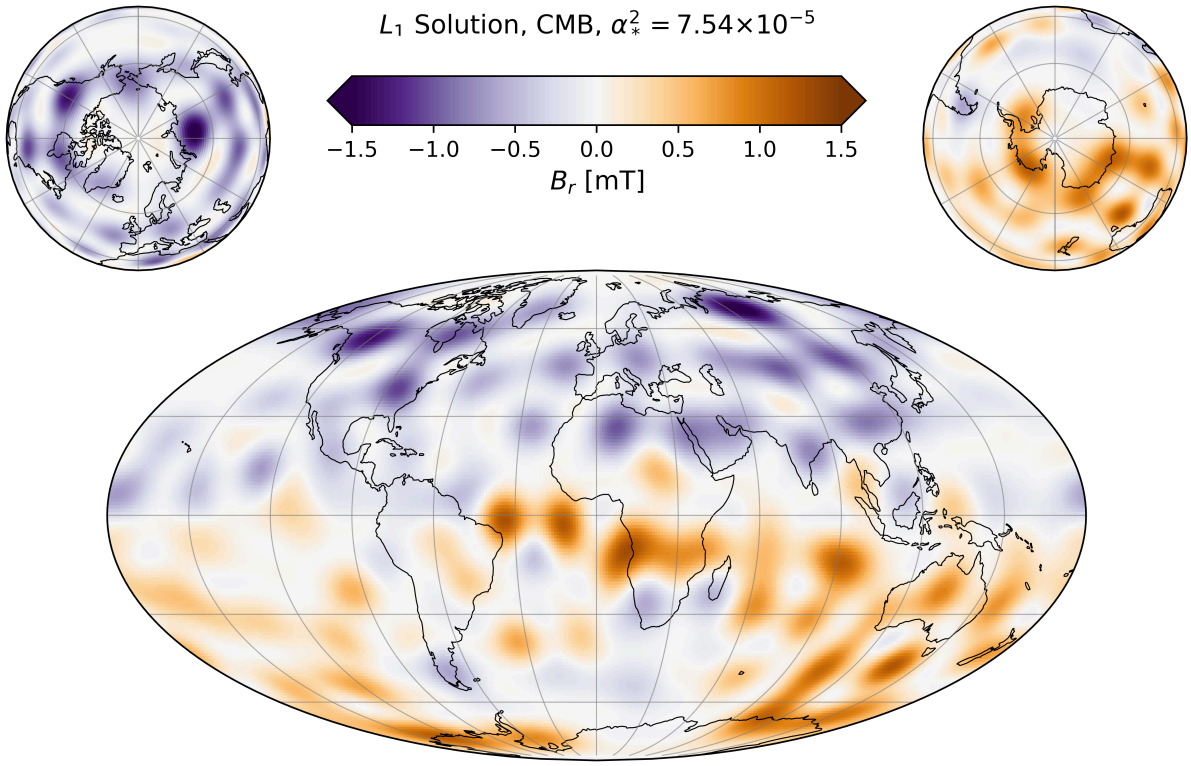


Figure 15: Large version of the plot shown in Figure 9b.

A.3 Implementation of Robust Iteratively Reweighted Least Squares Algorithm

```python
import numba
import numpy as np
import numpy.typing as npt
from typing import Protocol, Any
from tqdm.auto import trange

EPS = 1e-14  # Small value to prevent division by zero


@numba.njit
def huber_weights(residuals: npt.NDArray, c: float) -> npt.NDArray:
    denom = np.maximum(np.abs(residuals), EPS)
    return np.minimum(c / denom, np.ones_like(residuals))


def median_absolute_deviation_scale_estimate(x: npt.NDArray) -> float:
    CORRECTION_FACTOR = 1 / 0.675  # = 1/(Φ^(-1)(0.75))
    median = np.median(x)
    mad = np.median(np.abs(x - median))

    return max(CORRECTION_FACTOR * mad, EPS)


class MEstimator(Protocol):
    def __call__(self, scaled_residuals: npt.NDArray, /) -> npt.NDArray: ...


class ScaleEstimator(Protocol):
    def __call__(self, residuals: npt.NDArray, /) -> float: ...


class HuberWeights:
    def __init__(self, c: float = 1.345):
        self.c = c  # Cut-off

    @staticmethod
    @numba.njit
    def _inner(residuals: npt.NDArray, c: float) -> npt.NDArray:
        denom = np.maximum(np.abs(residuals), EPS)
        return np.minimum(c / denom, np.ones_like(residuals))

    def __call__(self, residuals: npt.NDArray) -> npt.NDArray:
        return self._inner(residuals, c=self.c)


@numba.njit
def laplace_weights(residuals: npt.NDArray) -> npt.NDArray:
    denom = np.maximum(np.abs(residuals), EPS)
    return 1 / denom


@numba.njit
def _get_regularisation_weights(gamma: npt.NDArray, p: float, delta: float):
    ekblom = (gamma**2 + delta**2) ** (p / 2 - 1)
    return ekblom

```

```
57
58  @numba.njit
59  def _construct_A(
60      G: npt.NDArray,
61      w: npt.NDArray,
62      H_k: tuple[npt.NDArray],
63      alpha_k: tuple[float],
64      delta_k: tuple[float],
65      p_k: tuple[float],
66      gamma_k: tuple[npt.NDArray],
67  ):
68      N_m = G.shape[1]
69      K = len(H_k)
70
71      reg_term = np.zeros((N_m, N_m))
72
73      for k in numba.literal_unroll(range(K)):
74          gamma = gamma_k[k]
75          p = p_k[k]
76          alpha = alpha_k[k]
77          delta = delta_k[k]
78          H = H_k[k]
79
80          w_m = _get_regularisation_weights(gamma, p, delta)
81          reg_term += (alpha**2) * H.T @ (w_m[:, None] * H)
82
83      GTWG = G.T @ (w[:, None] * G)
84      A = GTWG + reg_term
85
86      return A
87
88
89  def irls_lp_multi_robust(
90      G: npt.NDArray,  # Model
91      d: npt.NDArray,  # Data
92      H_k: tuple[npt.NDArray],  # Regularisers
93      alpha_k: tuple[float],  # Reg. Parameters
94      delta_k: tuple[float],  # Ekblom perturbations
95      p_k: tuple[float],  # Norm orders
96      robust_weighter: MEstimator = HuberWeights(),  # Robust weight function
97      scale_estimator: ScaleEstimator = median_absolute_deviation_scale_estimate,  # Scale estimator
98      N_iter: int = 1000,  # Maximum number of iterations
99      diff_threshold: float = 1e-6,  # Tolerance for early stopping
100     relative_diagonal_shift: float
101     | None = None,  # Diagonal shift to improve conditioning
102     return_iterates: bool = False,
103     progress: bool = False,
104     verbose: bool | int = False,
105     freeze_scaler_after: int
106     | None = None,  # Optionally freeze scale estimator after this many iterations
107     check_condition: bool = False,  # Compute condition number and report it in output
108     check_ekblom: bool = False,  # Optionally check and report Ekblom measures for debugging
109 ) -> tuple[
110     npt.NDArray,
111     npt.NDArray,
112     tuple[npt.NDArray],
113     list[dict[str, Any]] | None,
114     dict[str, Any],
```

```
]:
    """
    Solves a robust IRLS problem with an arbitrary number of regularisation terms
    and norm orders.

    Arguments:
        G:
            Forward/design matrix.
        d:
            Data vector (reshaped to 1D internally).
        H_k:
            Tuple of regularisation operators. Each operator maps model parameters
            to a regularisation domain for one term.
        alpha_k:
            Regularisation strengths for each term in H_k.
        delta_k:
            Ekblom perturbation constants for each regularisation term
            (stabilises LP-like weights).
        p_k:
            Norm orders for each regularisation term (for example 1.0 for L1-like behaviour).
        robust_weighter:
            Callable that maps scaled residuals to data weights.
        scale_estimator:
            Callable that estimates residual scale before robust weighting.
        N_iter:
            Maximum number of IRLS iterations.
        diff_threshold:
            Relative convergence threshold on model updates:
            ||m_new - m_old|| / (||m_new|| + eps).
        relative_diagonal_shift:
            Relative diagonal loading added to the system matrix each iteration:
            shift = relative_diagonal_shift * max(diag(A)).
            Helps numerical stability for ill-conditioned systems.
            Not used if None, otherwise it should be a small value like 1e-10 or smaller.
        return_iterates:
            If True, collect per-iteration diagnostics in iter_data.
        progress:
            If True, show a progress bar.
        verbose:
            If True, print per-iteration diagnostics.
            If it is an integer, only verbose every `i % verbose` iterations.
        freeze_scaler_after:
            If provided, update residual scale only for iterations i < freeze_scaler_after.
            After that, the most recent scale value is reused.
        check_condition:
            If True, compute and optionally warn about the condition number
            of the system matrix.
            Is expensive to calculate, so only use for debugging or if you suspect conditioning
            issues.
        check_ekblom:
            If True, compute and optionally report Ekblom measures for debugging.
    Returns:
        m: Estimated model vector.
        w: Final robust data weights.
        gamma_k: Final Ekblom measures.
        iter_data:
            Per-iteration diagnostics if return_iterates is True, otherwise None.
        info: Summary dictionary.
    """
```

```
173
174     ### Initialisation
175     N_d = G.shape[0]  # Number of data points
176     N_m = G.shape[1]  # Number of model parameters
177     K = len(H_k)  # Number of regularisation terms
178     N_k = tuple(H.shape[0] for H in H_k)  # Sizes of regularisation domains
179
180     ### Validation
181     d_vec = d.reshape(-1)
182     assert N_iter > 0
183     if freeze_scaler_after is not None:
184         assert freeze_scaler_after > 0
185     assert d_vec.shape == (N_d,)
186
187     assert len(alpha_k) == K
188     assert len(delta_k) == K
189     assert len(p_k) == K
190
191     ### Initialisation
192     m = np.zeros((N_m,))  # Model parameters
193     w = np.ones((N_d,))  # Data weights
194     gamma_k: tuple[npt.NDArray] = tuple(
195         np.zeros((N_k[k],)) for k in range(K)
196     )  # Ekblom measures
197
198     if return_iterates:
199         iter_data = []
200     else:
201         iter_data = None
202
203     if progress:
204         iter_ = trange(N_iter)
205     else:
206         iter_ = range(N_iter)
207
208     threshold_check = np.inf  # Initialize threshold check for convergence
209     residuals = np.zeros_like(np.inf)  # Initialize residuals
210     scale = 1.0
211
212     for i in iter_:
213         ### Setup system
214         A = _construct_A(G, w, H_k, alpha_k, delta_k, p_k, gamma_k)
215         b = G.T @ (w * d_vec)
216
217         ### Optionally shift diagonal to improve conditioning prior to solve
218         if relative_diagonal_shift is not None:
219             diag_shift = relative_diagonal_shift * np.max(np.diag(A))
220             A[np.diag_indices_from(A)] += diag_shift
221
222         ### Optional: Check condition number
223         if check_condition:
224             cond_number = np.linalg.cond(A)
225             if cond_number > 1e12:
226                 print(
227                     f"Warning: System is ill-conditioned at iteration {i}
                        (cond={cond_number:.2e}). Results may be unstable."
228                 )
229
230         ### Solve system
```

```
        # System is SPD, can use Cholesky for speedup
        m_new = scipy.linalg.solve(
            A, b, assume_a="pos"
        )  # Indicate that A is symmetric positive definite for speedup
        # m_new = scipy.linalg.solve(A, b)
        # m_new = np.linalg.solve(
        #     A, b,
        # )

        if not np.all(np.isfinite(m_new)):
            print(
                f"Non-finite values encountered in model update at iteration {i}. Terminating."
            )
            break

        ### Compute residuals and check convergence
        m_diff = m - m_new
        m_diff_L2 = np.linalg.norm(m_diff)

        m = m_new

        residuals = d_vec - G @ m

        ### Update robust weights
        if freeze_scaler_after is None or (
            freeze_scaler_after is not None and i < freeze_scaler_after
        ):
            scale = scale_estimator(residuals)

        w = robust_weighter(residuals / scale)

        ekblom_checks = []
        for k, H in enumerate(H_k):
            gamma_k[k][:] = H @ m

            if check_ekblom:
                ekblom_check = np.linalg.norm(gamma_k[k]) ** 2 / N_k[k]
                ekblom_checks.append(ekblom_check)

                if delta_k[k] ** 2 > (ekblom_check * 1e-3):
                    print(
                        f"Warning: Ekblom measure for regularisation term {k} is very small (||γ||^2/N={ekblom_check:.2e}), which may lead to numerical instability. Consider increasing delta_k[{k}]={delta_k[k]:.2e}."
                    )

        threshold_check = m_diff_L2 / (np.linalg.norm(m) + EPS)

        ### Debugging information
        if return_iterates:
            entry = {
                "i": i,
                "scale": scale,
                "threshold_check": threshold_check,
            }

            if check_condition:
                entry["cond"] = cond_number

```

```
            if check_ekblom:
                entry["ekblom_checks"] = ekblom_checks

            iter_data.append(entry)

        if verbose:
            if isinstance(verbose, int):
                if i % verbose != 0:
                    continue

            debug_msgs = [
                f"Iter {i}",
                f"m_diff_L2/||m||: {threshold_check:.2e}",
                f"max resid: {np.max(np.abs(residuals)):.2e}",
            ]
            if check_condition:
                debug_msgs.append(f"Condition number: {cond_number:.2e}")

            if check_ekblom:
                for k in range(K):
                    debug_msgs.append(
                        f"sqrt(||y_{k}||_2^2 / N_k): {np.sqrt(ekblom_checks[k]):.2e}"
                    )
            print(", ".join(debug_msgs))

        ### Terminate if converged
        if threshold_check < diff_threshold:
            break

    info = {
        "iterations": i + 1,
        "relative_norm": threshold_check,
        "residuals": residuals,
    }

    return m, w, gamma_k, iter_data, info
```